

1 How many distinct real solutions does the equation $x^3 - 3x^2 + 5 = 0$ have?

- A 0
- B 1
- C 2
- D 3

2 A circle has equation $x^2 + y^2 = 50$. A tangent to the circle at a point P is parallel to the line with equation $7x + y = 1$. The point P has positive x and y coordinates.

What are the coordinates of P ?

- A (1, 7)
- B (1, -7)
- C (5, 5)
- D (7, 1)
- E (7, -1)
- F (-7, -1)

3 A cubic polynomial $p(x)$ has stationary points at $x = 1$ and $x = 3$.

What is the value of $p(2)$?

- A -2
- B -1
- C 0
- D 2
- E 4
- F Cannot be determined from the information given.

- 4 A box contains 4 red balls, 5 blue balls, and 3 green balls. All 12 balls are identical apart from their colour.

Balls are drawn at random from the box, one at a time, without replacement.

- If a red ball is drawn, it is not replaced and another ball is drawn.
- As soon as a blue ball or a green ball is drawn, the process stops.

What is the probability that at least two red balls are drawn and the process stops on a blue ball?

- A $\frac{5}{132}$
- B $\frac{1}{22}$
- C $\frac{5}{88}$
- D $\frac{5}{72}$
- E $\frac{1}{11}$
- F $\frac{5}{8}$

- 5 Find the value of the constant k for which the equation

$$2 \cos^2(x) - 3 \sin(x) = k$$

has exactly three distinct solutions in the interval $0 \leq x < 2\pi$.

- A $k = -3$
- B $k = -1$
- C $k = 2$
- D $k = \frac{25}{8}$
- E $k = 3$
- F $k = 5$

- 6 Variables x and y are related by the equation $y = Ax^n$, where A and n are constants. A graph of $\log_2 y$ plotted against $\log_2 x$ is a straight line with gradient 2 and intercept 4 on the vertical axis.

Which of the following expresses y in terms of x ?

- A $y = 4x^2$
- B $y = 16x^2$
- C $y = 2x^4$
- D $y = 4x^4$
- E $y = 2^{2x+4}$

- 7 Consider the family of quadratic curves given by

$$y = x^2 - 2ax - 4a + 5$$

where a is a real constant.

What is the complete range of values of a for which the minimum point of the curve lies strictly in the second quadrant (that is, where $x < 0$ and $y > 0$)?

- A** There are no such values of a
- B** $a < -5$
- C** $-5 < a < 0$
- D** $-5 < a < 1$
- E** $0 < a < 1$
- F** $a < -5$ or $a > 1$
- G** $a < -5$ or $0 < a < 1$
- 8 The sum of the first 10 terms of a sequence is 200. Each term differs from the previous term by a constant amount. The sum of the next 10 terms is 600.
What is the value of the first term?

- A** 0
- B** 1
- C** 2
- D** 4
- E** 11

- 9 A geometric series has first term a and common ratio r . All terms in the series are non-zero, and the sum to infinity of the series exists.

The sum of the first two terms is equal to $\frac{5}{9}$ of the sum to infinity. The terms in the series do not all have the same sign.

What is the third term of the series, expressed as a fraction of the sum to infinity?

- A** $\frac{4}{27}$
- B** $\frac{19}{27}$
- C** $\frac{20}{27}$
- D** $\frac{4}{9}$
- E** $\frac{35}{27}$
- F** $\frac{-10}{9}$

- 10** Variables x and y are related by the equation $y = kx^n$, where k and n are constants.
A graph of $\log_{10} y$ against $\log_{10} x$ is a straight line passing through the points $(0, 2)$ and $(4, 0)$.
Which of the following expresses y in terms of x ?

A $y = 2 - \frac{x}{2}$

B $y = \frac{2}{\sqrt{x}}$

C $y = \frac{2}{x^2}$

D $y = \frac{100}{\sqrt{x}}$

E $y = \frac{100}{x^2}$

- 11** The function $f(x) = x^3 - 3x^2 - 9x + 30$ is defined on the restricted domain $-2 \leq x \leq 2$.
What is the range of $f(x)$?

A $3 \leq f(x) \leq 28$

B $3 \leq f(x) \leq 35$

C $8 \leq f(x) \leq 28$

D $8 \leq f(x) \leq 35$

E $-\infty < f(x) < \infty$

- 12** What is the coefficient of x^4 in the expansion of $(1 + 2x + 2x^2)^5$?

A 120

B 210

C 240

D 320

E 360

- 13** Consider the equation

$$\tan\left(\frac{4\pi}{x}\right) = \sqrt{3}$$

for $x > 0$.

The positive solutions are arranged in decreasing order, $x_1 > x_2 > x_3 > \dots$

What is the value of x_3 ?

- A** $\frac{6}{5}$
 - B** $\frac{12}{13}$
 - C** $\frac{12}{7}$
 - D** $\frac{24}{13}$
 - E** 3
 - F** 12
- 14** A sector of a circle has a fixed perimeter of 60.
What is the radius of the sector that maximises its area?
- A** $30/\pi$
 - B** $60/(\pi + 2)$
 - C** 15
 - D** 20
 - E** 30
- 15** The positive numbers α and β satisfy the relation $\alpha = 3\beta$. A point P on the y -axis is a distance of 5 from the point $(\alpha, 0)$ and a distance of 3 from the point $(\beta, 0)$.
What is the value of the product $\alpha\beta$?
- A** 2
 - B** 6
 - C** 7
 - D** 16
 - E** 18
 - F** 24

- 16** A solid right circular cylinder has base radius r and height h , as shown in the diagram. The base radius r is decreased by 20%, while the volume of the cylinder is increased by 28%. By what percentage does the height h increase?

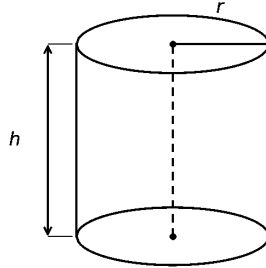


Diagram not to scale

- A** 48%
B 50%
C 60%
D 68%
E 100%
F 200%
- 17** A sequence is defined by

$$u_1 = 2, \quad u_{n+1} = \frac{1}{1 - u_n}.$$

What is the value of u_{2026} ?

- A** -2
B -1
C $-\frac{1}{2}$
D $\frac{1}{2}$
E 2

- 18 Find the area of the finite region in the first quadrant bounded by the curve $y = 2x^2$, the line $y = 32$, and the y -axis.

- A $\frac{128}{3}$
 B 64
 C $\frac{256}{3}$
 D 128
 E $\frac{320}{3}$
 F $\frac{512}{3}$

- 19 $ABCD$ is a trapezium with parallel sides AB and CD . The diagonals AC and BD intersect at point X , as shown in the diagram.

$AX = 1$ cm, $AB = x$ cm, $CX = (x + 2)$ cm, and $CD = (4x + 1)$ cm.

What is the length, in cm, of CD ?

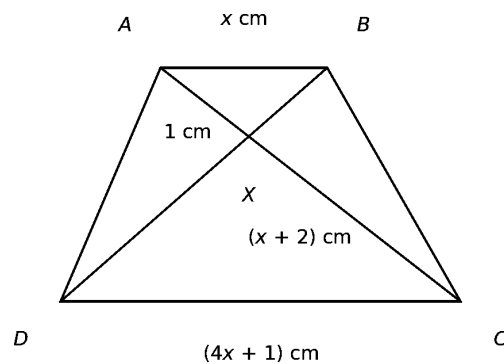


Diagram not to scale

- A $3 + 2\sqrt{5}$
 B $4 + \sqrt{2}$
 C $4 + 4\sqrt{2}$
 D $5 + 4\sqrt{2}$
 E $5 + 4\sqrt{3}$
 F $5 + 8\sqrt{2}$

- 20** The polynomial $(x + y)^{37}$ is expanded in ascending powers of x .
How many of the terms in this expansion have a coefficient that is divisible by 37?
- A** 35
B 36
C 37
D 38
E 0
- 21** Which one of the following statements describes a mathematically impossible scenario for a polynomial $p(x)$ with real coefficients?
- A** A quadratic polynomial $p(x)$ has two distinct positive real roots and its derivative at $x = 0$ is positive.
B A cubic polynomial $p(x)$ has a local maximum at $x = 1$, a local minimum at $x = 4$, and $p(1) < p(4)$.
C A cubic polynomial $p(x)$ has exactly one real root, and its derivative $p'(x)$ has two distinct real roots.
D A cubic polynomial $p(x)$ has no stationary points and passes through the origin.
E A quartic polynomial $p(x)$ has exactly one stationary point.
F A cubic polynomial $p(x)$ has a local minimum at $x = 1$ and a local maximum at $x = 4$.
- 22** A box contains 12 electronic components, of which 4 are defective and 8 are working. Components are drawn at random from the box, one at a time without replacement, and tested sequentially.
Testing stops immediately once a second defective component has been identified.
What is the probability that fewer working components than defective components have been tested when the process stops?
- A** $\frac{1}{11}$
B $\frac{8}{55}$
C $\frac{9}{55}$
D $\frac{13}{55}$
E $\frac{17}{55}$
F $\frac{23}{55}$

- 23** Functions f and g are defined by

$$f(x) = x^2 - 1, \quad g(x) = 2x + 1.$$

The equation

$$f(g(x)) = g(f(x))$$

has two real solutions p and q . What is the value of $p + q$?

- A** -4
- B** -2
- C** -1
- D** 1
- E** 2

- 24** The function $f(x)$ is defined by

$$f(x) = \frac{5^x}{5^x + \sqrt{5}}$$

What is the value of the following sum?

$$S = \sum_{r=0}^{100} f\left(\frac{r}{100}\right)$$

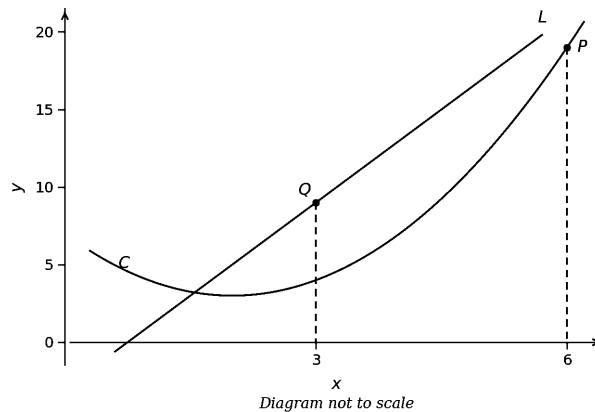
- A** 49.5
- B** 50
- C** 50.5
- D** 51
- E** 100
- F** 101

- 25** The diagram shows part of the curve C with equation $y = (x - 2)^2 + 3$ and part of the straight line L with equation $y = 4x + c$, where c is a constant.

The point P lies on C at $x = 6$, and the point Q lies on L at $x = 3$.

The y -coordinate of P is equal to twice the y -coordinate of Q plus 1.

What is the y -coordinate of the point on L with x -coordinate 5?



- A** -3
B 9
C 17
D 17.5
E 23
F 25
G 47
- 26** A right circular cone has base radius r and perpendicular height h . The volume of the cone is V and its total surface area is A . These quantities are related by the equation:

$$\ln(9V) - \ln(A) = \ln(r)$$

Which of the following expresses h in terms of r ?

- A** $h = \frac{3}{4}r$
B $h = \frac{1}{2}r$
C $h = \frac{3}{8}r$
D $h = \frac{4}{3}r$
E $h = \frac{9}{40}r$
F $h = \frac{\sqrt{2}}{4}r$

27 Let P and Q be positive real numbers.

Q is the value obtained after P is first increased by 100%, then the result is decreased by 75%, and finally that result is increased by 100%.

What is the value of $\log_4(Q) - \log_4(P)$?

- A** -1
- B** $-\frac{1}{2}$
- C** 0
- D** $\frac{1}{2}$
- E** 1